



**Thailand Mathematical Olympiad 2023**

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– Day 1

**1** Let  $A$  be set of 20 consecutive positive integers, Which sum and product of elements in  $A$  not divisible by 23. Prove that product of elements in  $A$  is not perfect square

**2** Let  $\triangle ABC$  which  $\angle ABC$  are right angle, Let  $D$  be point on  $AB$  ( $D \neq A, B$ ), Let  $E$  be point on line  $AB$  which  $B$  is the midpoint of  $DE$ , Let  $I$  be incenter of  $\triangle ACE$  and  $J$  be  $A$ -excenter of  $\triangle ACD$ . Prove that perpendicular bisector of  $BC$  bisects  $IJ$

**3** Defined all  $f : \mathbb{R} \rightarrow \mathbb{R}$  that satisfied equation

$$f(x)f(y)f(x-y) = x^2f(y) - y^2f(x)$$

for all  $x, y \in \mathbb{R}$

**4** A table tennis tournament has 101 contestants, where each pair of contestants will play each other exactly once. In each match, the player who gets 11 points first is the winner, and the other the loser. At the end of the tournament, it turns out that there exist matches with scores 11 to 0 and 11 to 10. Show that there exists 3 contestants  $A, B, C$  such that the score of the losers in the matches between  $A, B$  and  $A, C$  are equal, but different from the score of the loser in the match between  $B, C$ .

**5** Let  $\ell$  be a line in the plane and let  $90^\circ < \theta < 180^\circ$ . Consider any distinct points  $P, Q, R$  that satisfy the following:

(i)  $P$  lies on  $\ell$  and  $PQ$  is perpendicular to  $\ell$

(ii)  $R$  lies on the same side of  $\ell$  as  $Q$ , and  $R$  doesn't lie on  $\ell$

(iii) for any points  $A, B$  on  $\ell$ , if  $\angle ARB = \theta$  then  $\angle AQB \geq \theta$ .

Find the minimum value of  $\angle PQR$ .

– Day 2

**6** Let  $a, b, c, x, y$  be positive real numbers such that  $abc = 1$ . Prove that

$$\frac{a^5}{xc+yb} + \frac{b^5}{xa+yc} + \frac{c^5}{xb+ya} \geq \frac{9}{(x+y)(a^2+b^2+c^2)}.$$

- 7 Let  $n$  be positive integer and  $S = 0, 1, \dots, n$ , Define set of point in the plane.

$$A = \{(x, y) \in S \times S \mid -1 \leq x - y \leq 1\}$$

, We want to place a electricity post on a point in  $A$  such that each electricity post can shine in radius 1.01 unit. Define minimum number of electricity post such that every point in  $A$  is in shine area

- 8 Let  $ABC$  be an acute triangle. The tangent at  $A, B$  of the circumcircle of  $ABC$  intersect at  $T$ . Line  $CT$  meets side  $AB$  at  $D$ . Denote by  $\Gamma_1, \Gamma_2$  the circumcircle of triangle  $CAD$ , and the circumcircle of triangle  $CBD$ , respectively. Let line  $TA$  meet  $\Gamma_1$  again at  $E$  and line  $TB$  meet  $\Gamma_2$  again at  $F$ . Line  $EF$  intersects sides  $AC, BC$  at  $P, Q$ , respectively. Prove that  $EF = PQ + AB$ .

- 9 Prove that there exists an infinite sequence of positive integers  $a_1, a_2, a_3, \dots$  such that for any positive integer  $k$ ,  $a_k^2 + a_k + 2023$  has at least  $k$  distinct positive divisors.

- 10 To celebrate the 20th Thailand Mathematical Olympiad (TMO), Ratchasima Witthayalai School put up flags around the Thao Suranari Monument so that

- Each flag is painted in exactly one color, and at least 2 distinct colors are used.
- The number of flags are odd.
- Every flags are on a regular polygon such that each vertex has one flag.
- Every flags with the same color are on a regular polygon.

Prove that there are at least 3 colors with the same amount of flags.